

Male Genital Lichen Sclerosus

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Abstract

Male genital lichen sclerosus (MGLSc) is a chronic inflammatory skin disease responsible for male sexual dyspareunia and urological morbidity. An afeared complication is squamous cell carcinoma (SCC) of the penis. The precise etiopathogenesis of MGLSc remains controversial although genetic, autoimmune and infective (such as human papillomavirus (HPV) hepatitis C (HCV), Epstein-Barr virus (EBV) and Borrelia) factors have been implicated: Consideration of all the evidence suggests that chronic exposure of susceptible epithelium to urinary occlusion by the foreskin seems the most likely pathomechanism. The mainstay of treatment is topical ultrapotent corticosteroid therapy. Surgery is indicated for cases unresponsive to topical corticosteroid therapy, phimosis, meatal stenosis, urethral stricture, carcinoma *in situ* (CIS) and squamous cell carcinoma.

Key Words: Male genital lichen sclerosus, etiology, pathogenesis, management

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What was known?

MGLSc is a cause of male sexual dysfunction and urological morbidity and associated with squamous carcinoma of the penis. It has been held to be an autoimmune disease.

Introduction

MGLSc is an acquired, chronic inflammatory cutaneous disease of disputed etiology. It causes acute and chronic balanoposthitis and scarring that can result in significant sexual and urinary dysfunction as well as conferring a risk of squamous carcinoma of the penis.^[1,2]

The exact incidence and prevalence are difficult to ascertain. This could be because it is managed by many specialties (dermatology, genito-urinary medicine, urology and pediatrics), as well as significant under-recognition and hence under-reporting by patients and physicians. The estimated incidence of LSc ranges from 1/300 (0.3%) to 1/1000 (0.1%) in both sexes.^[3] A more recent study reported the incidence of MGLSc of 1.4 patients per 100, 000 visit.^[4] The epidemiology may vary between countries and racial groups because MGLSc is principally a disease of the uncircumcised although it can persist or recur after circumcision. There is a bimodal age incidence with peaks in young boys and in adult men, late in the fourth decade.^[4-8]

MGLSc can be the cause of significant morbidity. The symptomatology is an expression of preputial and urethral dysfunction very often summarized as male

dyspareunia.^[2,6,9] The spectrum of presentation is wide. It may be asymptomatic. Some patients describe spontaneous itching, burning, bleeding, tearing, splitting (of the prepuce and frenulum), or hemorrhagic blisters or urinary problems such as dysuria and narrowing of the urinary stream, or anatomical changes of the genitalia. Most men (55%) have symptoms associated with sexual activity but dyspareunia may be even more common than this statistic due to failure or difficulty in eliciting the symptomatology.^[6] Other presentations are non-retractile foreskin (phimosis), foreskin fixed in retraction (paraphimosis), urethritis, urethral stricture, urinary retention and renal failure. Some cases are diagnosed at the presentation of CIS or frank SCC. The disease and its complications have a significant impact on male sexual health.^[6,10]

Signs may be subtle or florid. The organ should be examined systematically and sedulously, including with a hand lens. The classical genital morphologic manifestations of MGLSc are similar to those features of the disease encountered at extragenital sites, and include atrophic ivory-white patches (leukoderma) or plaques, hypertrophic, slightly scaly patches or plaques with telangiectasia and sparse purpura. A constrictive lichenoid posthitis is commonly seen associated with a fibrotic preputial band causing "hourglass" "waisting" of the penile shaft. Variable asymmetric, sometimes florid, Zoonoid inflammation may be present and distract the clinician from the correct diagnosis. The glans may be etiolated, affected by sclerosis especially parameatally and show purpura, ecchymosis or angiokeratomas. Meatal involvement may be subtle with encroachment

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and blunting of the architecture or more overt with “pin hole” narrowing of the meatus, but may be more overt and extensive. Adhesions occur and can be subcoronal or transcoronal. Anatomical definition is lost with effacement of the coronal sulcus and early penile papules. The frenulum is a particular target manifesting sclerosis or complete obliteration of the frenulum.^[1,2,6] Perianal involvement with MGLSc in men is very rare.^[1,2] Concomitant extragenital LSc is also extremely rare (<0.1%).^[6,10]

Etiology

Genetic factors

Genetic factors have been proposed on the basis on reports of familial cases and association with HLA antigens. Familial LSc has been observed in identical and non-identical twins, sisters, mothers and daughters.^[11,12] Family history was found in 12% of female genital (FG) LSc.^[11] However, there is little evidence of a familial predisposition to LSc in boys.^[11,13] The human leukocyte antigen (HLA) complex is known to determine an individual's susceptibility to inflammatory diseases by influencing both cellular and humoral immunity. Majority of this work involved women. Several studies have found HLA DQ7 and, to a lesser extent, HLA DQ8 and DQ9 to be more common in FG LSc than in controls,^[14] especially if the onset of LSc is in childhood.^[15] HLA-B*08, B*18, -B*15, B*57, -CW*03, -CW*07, CW*18, -DQ3, -DQ7, -DRB1*04, -DRB*4, -DRB1*07, -DRB1*12 and the haplotypes HLA-DRB1*12, -DQB1*03:01/04/09/010 were found to increase susceptibility to FG LSc, whereas HLA-DR17, -DRB1*03:01/04 and the haplotypes HLA-DRB1*03, -DQB1*02, -DRB1*03:01, DQB1*02:01/02/03 appear to protect against (FG) LSc.^[12,16,17] Increased frequencies of HLA-DR11, -DR12 and DQ7 have been reported in MGLSc.^[18]

GLSc has been associated with atopic eczema in boys.^[19] Filaggrin (FLG) is a key protein that facilitates terminal differentiation of the epidermis and formation of the epithelial barrier, and mutations have been found to be present in atopic dermatitis. A barrier defect due to *FLG* mutations might contribute to the susceptibility of genital epithelium to urinary irritation in MGLSc.^[20] However, no significant differences were found in a study of 93 MGLSc with the most prevalent filaggrin loss-of-function mutations (R501X, 2282del4, R2447X and S3247X).^[21]

Autoimmunity

It remains debatable whether autoimmunity underlies this condition. A personal and/or family history of autoimmunity has been shown to be associated with LSc. Organ-specific antibodies such as anti-thyroid, gastric (parietal-cell) tissue, intrinsic factor, antinuclear and anti-smooth muscle autoantibodies

have been found in patients with LSc. Autoimmune conditions such as thyroid disease, diabetes, vitiligo, alopecia areata, pernicious anemia, scleroderma and rheumatoid arthritis have been reported in patients with LSc.^[12,15,22-25] The similarities of some clinical and histopathological features in LSc and lipid proteinosis, a rare autosomal recessive genodermatosis associated with pathogenic loss-of-function mutations in the extracellular matrix protein 1 (ECM1) gene, has led to the hypothesis that ECM1 autoimmunity might contribute to the etiopathogenesis of LSc.^[26,27] Further study suggested although autoantibodies to ECM1 have been found in both FG- and MGLSc, but this may be an epiphenomenon.^[27,28] Similar conclusion has also been drawn for the role of circulating anti 180NC 16a and anti-BP230 autoantibodies because the percentage of patients with positive circulating anti-BP 180 antibodies appears not to exceed the number found in general population.^[29,30] It has been proposed there might be different pathogenetic background in male and female patients with preponderance of the role of autoimmunity in (FG) LSc.^[31] Furthermore, different percentages of CD4+ and FOXP3+ lymphocytes were observed in (FG)- and MGLSc suggesting a different pathogenetic pathway.^[32]

Infection

- (i) Acid-fast bacilli and spirochetes
Previous studies have linked several infective agents with LSc. These include acid-fast bacilli and spirochetes.^[33-35]
- (ii) *Borrelia burgdorferi*
Acrodermatitis chronica atrophicans caused by *Borrelia burgdorferi* has some clinical and histological similarities with LSc. Many studies to date have produced conflicting results, which could be due to regional variations in *Borrelia burgdorferi* and different experimental technique used.^[36,37] Such an association has been negated.^[38]
- (iii) HPV
Human papillomavirus (HPV) infection has been implicated as a causative agent. A variety of HPV sub-types such as HPV 16, 18, 33 and 51 have been reported in MGLSc.^[39] It remains debatable whether some of these HPV are merely skin surface “passenger HPV” or “driver HPV” that are implicated in the pathogenesis of MGLSc. p16^{INK4a} can be used as a surrogate biomarker to confirm the diagnosis of HPV-infected penile lesions.^[39-42] Recent evidence has shown that there is a lack of clinical correlation of MGLSc with HPV and MGLSc has an HPV-unrelated transcriptome,^[6,43] endorsing the view that HPV might be an innocent bystander.^[43] Another study has shown that there was no difference in prevalence of MGLSc between HPV-positive and -negative samples.^[44] HPV has also shown to

be absent from the foreskin of boys circumcised for persistent phimosis after their first year of life and that topical glucocorticoid treatment failure is not associated with HPV or any specific histopathological changes.^[45] However, strong immunostaining for p16^{INK4a} correlated with HPV16/18 infection in 6 out of 18 cases of MGLSc has been reported.^[39] A larger study is required to examine this association. Recent work has shown that there is no correlation of MGLSc ($n = 62$; HPV positivity = 37.9%) with p16^{INK4a} immunostaining ($P = 0.53$) suggesting the detection of skin surface “passenger HPV” and it is unlikely to play a role in the etiopathogenesis of MGLSc.^[46]

(iv) HCV

HCV has been implicated by case reports^[47,48] but a recent study from our group ($n = 61$) showed that not one patient with MGLSc had HCV suggesting that it is highly unlikely to play a pathogenic role in MGLSc.^[49]

(v) EBV

Epstein-Barr virus (EBV) DNA has been found in 26.5% of 34 vulvar biopsies with FGLS compared with 0% in controls^[50] but its role in MGLSc is not known.

Environmental factors

LSc demonstrates the Koebner phenomenon. It arises from areas that have undergone trauma, in old scars (e.g. after vulvectomy and circumcision), on skin grafts, at sites prone to constant friction, and after sunburn or radiation treatment.^[51-56]

Exposure to urine

MGLSc is unequivocally a disease of the uncircumcised male.^[6] MGLSc is exceedingly rare in the male circumcised at birth indicating that the foreskin must play an obligate role in the etiopathogenesis of MGLSc. However, MGLSc does occur in the circumcised male: with hypospadias, with genital jewelry (piercing), after surgery, trauma and instrumentation and in the bariatric patient. It does recur in grafts especially skin grafts more than mucosal grafts.^[6,57-59]

Post-micturition dribbling or microincontinence has been proposed to play a central role in the etiopathogenesis of MGLSc.^[60-64] Many men presenting with MGLSc confess to dribbling post-micturition. A recent study by our group has shown that 91-100% men with MGLSc have dribbling compared to 14% controls implicating the role of post-micturition microincontinence in MGLSc.^[65] Furthermore, MGLSc patients are often found to have an abnormal urethral meatus or navicular fossa on meticulous examination. In the circumcised male, a tiny drop of urine appearing at the meatus post-micturition will have negligible contact with a keratinized glans before being absorbed by undergarments. In an uncircumcised male, with a similarly dysfunctional terminal urethral apparatus,

the situation is very different. In men, dribbled urine, consequent upon post micturition microincontinence, is likely to pool and become occluded between the inner prepuce and distal penis/glans, affecting the frenulum, perimeatal glans and visceral prepuce. Occlusion and the phenomenon of koebnerization precipitate inflammation; inflammation progresses to sclerosis.^[6,61,63,64,66] The urinary hypothesis is also supported by the observation of LSc occurring around perineal urethrostomies in patients who undergone perineal urethroplasty for long urethral strictures or severely scarred urethral plate. Chronic intermittent leakage of urine from the perineal urethrostomy following voiding is inevitably occluded between juxtaposed perineal folds with identical pathological consequences in the affected skin to the penile situation described above.^[67] This hypothesis is strengthened by the occurrence of extragenital LSc in the peristomal skin in urostomy patients likely due to occlusion and urine contact.^[68]

Whether MGLSc is a non-specific pathological response to urinary exposure or there is some specific mechanism or culpable constituent of urine remains unknown. Magnetic resonance spectroscopy study suggests that there is not one single indictable component of urine.^[69]

Other factors

(i) Drugs

A small study has shown an inverse relationship between the presence of vulval LS and use of beta-blockers and ACE inhibitors. ACE inhibitors may diminish the LSc inflammatory process and therefore protect from LSc. Beta-blockers block cyclic AMP level resulting in upregulation of keratinocyte proliferation, reduced differentiation and increased motility of lymphocytes, which may influence the clinical expression of patients with LSc.^[70] The role of drugs in the causation of MGLSc is not known.

(ii) Metabolic disease

A recent case-controlled study suggested that there might be a metabolic or lifestyle component in the etiology of MGLSc. Increased rates of elevated body mass index, diabetes mellitus, coronary artery disease (CAD) and smoking were observed in MGLSc compared to control group.^[71,72] Diabetes mellitus and CAD are known to be associated with microvascular disease systematically, and the authors suggested an association of LSc with direct vascular compromise.^[71] However, obese men are frequently seen with GLSc, including those where suprapubic obesity and retraction of the penis (the “vanishing penis”) into the pubic mound create a pseudo or neo foreskin in males circumcised at birth or later.

MGLSc and SCC

The risk of transformation of MGLSc to carcinoma *in situ* (CIS) is difficult to quantify. Early microinvasive

disease might be challenging to diagnose clinically against a background of LSc. A retrospective review by Barbagli *et al.* of 130 patients with MGLSc, revealed 11 men (8.4%) with premalignant or malignant features (7 cases with SCC, 2 cases with verrucous carcinoma, 1 case of SCC associated with verrucous carcinoma and 1 case of erythroplasia of Queyrat).^[73] The time interval between diagnosis of LSc and the development of SCC was 14-30 years. Two forms of penile intraepithelial neoplasia (PeIN) have been described (analogy with the vulvar intraepithelial neoplasia classification): Differentiated PeIN which is often associated with MGLSc and undifferentiated or Bowenoid PeIN which is associated with HPV infection.^[74]

The association of GLSc and SCC is widely recognized and the risk has been estimated at between 2% and 12.5%.^[13,73,75-77] MGLSc is also considered one of the two pathways for the development of CIS and penile cancer, the other being via HPV infection.^[2,74] However, in a large series of patients from our clinics not one patient developed SCC, suggesting that accurate diagnosis and definitive curative management are central to arresting carcinogenic drive.

MGLSc and melanocytic lesions

Atypical genital melanocytic naevus has been reported concomitantly in patients with FG- and MGLSc.^[78-80] Cases of melanoma complicating FGLSc have appeared in the literature.^[81-85] We have seen three in MGLSc.^[86] It has been postulated that LSc might be an immune response possibly triggered by the melanocytic lesion, alternatively that the inflammatory environment of LSc provides a microenvironmental niche promoting melanocytic neoplasia.

Diagnosis and Treatment

Most cases of MGLSc can be diagnosed clinically. Under some circumstances a biopsy can be performed. A biopsy is useful to exclude neoplastic transformation and differentiate LSc from erosive lichen planus, scarring cicatricial pemphigoid and lichen simplex.^[1,2] The histology of LSc is characterized by flattening of the epidermis with variable hyperkeratosis, hydropic degeneration of basal cells and an inflammatory cell dermal infiltrate. However, biopsy histology of MGLSc may be non-specific, depending on timing, site and severity.^[87] A biopsy of the most clinically abnormal area, for example, of inflammation or scarring, may not yield a histological diagnosis. There may be nonspecific inflammatory changes (e.g. zoonoid) leading to an erroneous diagnosis of Zoon's balanitis. In many men the preputial or glanular changes may be secondary and reactive to the balanopreputial dysfunction created by the principal proximate or adjacent or apposed site of the LSc. A negative biopsy does not exclude LSc. A negative biopsy does not exclude neoplasia.^[2,9,66,88]

The treatment of MGLSc aims to achieve optimal symptomatic relief, prevention of further deterioration of anatomical architecture changes and a theoretical mitigation of malignant transformation. Our understanding of the role of occlusional contact with urine in the pathogenesis is fundamental to the effective management and the prognosis of MGLSc.^[65,89]

Treatment protocols in GLSc are well-established (level evidence III).^[2,6,66,90] Avoidance of contact with soap and urine are essential. Regular use of skin barrier emollient should be instituted. Short, trimmed pubic hair is recommended to avoid hair trapping between the foreskin and the glans penis and hence minimize irritation, abrasion and consequent inflammation.^[89] Early aggressive treatment may prevent disease progression.^[91] An ultrapotent topical corticosteroid (usually clobetasol propionate (level evidence I)) used under supervision for a finite course is effective^[2,92-94] and usually problem free, but herpes simplex and wart reactivation do occur.^[90,95] prophylactic aciclovir should be prescribed for patients with a history of genital HSV. Most men achieved remission by ultrapotent topical corticosteroid (50-60%).^[6] Topical calcineurin inhibitors should not be used in MGLSc because of a theoretical synergistic risk of SCC.^[6,61,62,69] If maximal conventional medical treatment is not possible or fails then circumcision is recommended. Removal of the foreskin alters the local environment of glans penis: It removes the occlusive and koebnerizing effects of the foreskin and hence mitigates the consequences of post-micturition microincontinence. Circumcision is curatively effective in the majority (at least > 75% but precise long term follow up data are lacking).^[6] Some men might require advanced urethromeatal surgery. Our results show that accurate diagnosis and rational management restores sexual function and mitigates the risk of urinary dysfunction and can conserve the foreskin in about 50%. Furthermore, not one patient of the many hundreds of patients seen in the clinic over 20 years has yet developed penis cancer.^[6]

Conclusion

MGLSc is an acquired, chronic, inflammatory and fibrosing cutaneous disease responsible for significant sexual and urological dysfunction as well as creating a risk of squamous carcinoma of the penis. The etiology of MGLSc is now much better understood. It is not primarily an autoimmune condition. It is likely that an interaction between the irritant effects of urine and other pathogenic factors, such as chronicity, occlusion and as yet undetermined differential epithelial susceptibility, or reaction, to injury, are necessary for the development of MGLSc. Better understanding the pathogenesis of MGLSc is important to minimize sexual morbidity, urological dysfunction, mitigate the cancer risk it generates and preserve the foreskin when possible.

What is new?

MGLSc is due to chronic urinary micro incontinence and occlusion. It is treated with either a short course of ultrapotent topical steroid or circumcision.

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